
Summary

- Analytical scientist with expertise in method development/validation for small molecules and quality management in analytical laboratories.
- 8+ years of mass spectrometry (LC MS) experience.
- Proficient with QqQ (Waters TQD and TQS), Orbitrap (Q Exactive and Orbitrap Elite), QToF (Synapt G2Si and AbSciex x500R) and MALDI Imaging with Synapt G2Si.
- Fields of application include clinical metabolomics/lipidomics, preclinical drug discovery, feed and food safety, and environmental analysis.
- Implemented testing methods as per FDA bioanalytical validation and EC 657/2002 in ISO 17025:2005 accredited high throughput testing laboratory for official analysis.
- Collaborated with scientists from various disciplines. Contributed to various conferences through talks and posters, published in peer reviewed journals. Involved in teaching assistance for classes and teaching of incoming students in the laboratory. Supported in preparation of research grant applications.

Experience

September 2017 – Current

Research Fellow

Pharmacokinetics Core

University of Michigan, Ann Arbor, USA

Responsibilities:

- Quantitative method development, validation as per FDA bioanalytical validation and analysis of small molecule drug candidates in the scope of pharmacokinetics/dynamics studies using LC – MS instruments (AB Sciex QqQ and QToF, Waters TQD, Synapt G2 Si).
- Development of a method for Bile acids in clinical samples.
- Mass spectrometry imaging of small molecules using Synapt G2 Si.

April 2015 – June 2017

Postdoctoral Research Scientist

Beaumont Research Institute

William Beaumont Health System, Royal Oak, MI, USA

Responsibilities:

- Development/set up of targeted/untargeted methodologies and workflows for metabolomics and lipidomics using LC-HRMS.
- Management and maintenance of LC-MS instruments (LTQ Orbitrap Elite, Waters TQ-S) and workflows.
- Identification of unknown metabolites.
- Organize and manage programs of research in the LC MS laboratory.
- Worked in the field of clinical metabolomics applied to Obstetrics/Gynaecology and Neurology projects.
- Dissemination of research findings through publications and preparation of grant proposals.
- Training incoming students.

August 2010 – February 2013

PhD: Research Stay

Laboratory of Public Health Agency of Barcelona (ASPB)
Spain

Responsibilities:

- Development of methodologies for analysing veterinary drugs in food (honey, animal kidney, urine, muscle) and feed samples using various sample preparation techniques, LC-QqQ and LC-Q Orbitrap.
- Validate methods according to EC 657/2002 and implement for routine analysis according to ISO17025:2005.
- Support in routine maintenance of the instruments. Preparation of Standard Operating Procedures (SOP) and validation plans.

February – May 2013

PhD: Research Stay

Swiss Federal Institute of Aquatic Science and Technology (EAWAG)
Switzerland

Responsibilities:

- Development of experimental and data analysis workflow for non target and suspect screening of environmental samples using LC - Q Orbitrap
- Use of various data analysis tools – Envimass, MetFrag, Chemspider, Molgen MS/MS and ToX ID.

Education/Internships

May 2010 – April 2014

PhD in Analytical Chemistry

Methodologies for the analysis of veterinary drugs and growth promoters in the scope of food safety control.
University of Barcelona, Spain.

September 2008 – February 2010

Erasmus Mundus masters in Quality in Analytical Labs (EMQAL)

Food testing, data management and quality control systems in analytical laboratories.
University of Algarve, Portugal and University of Barcelona, Spain.

February – August 2008

Internship in Knowledge Process Outsourcing (KPO)

Scope e-Knowledge Center, India.

Responsibilities: Summarizing industrial patents into abstracts.

May – September 2007

Shift Supervisor

Vinkem Labs Pvt. Ltd., India.

Responsibilities: Supervision of pharmaceutical production process, Handling and maintenance of chemical reactors and chemical process equipments.

December 2006 – March 2007

Research project work

University of Madras, India.

Title: Production of edible films incorporated with antimicrobial enzymes for preservation of carrots.

July 2004 – May 2007

Bachelor of Technology in Biotechnology

SASTRA University, India.

Awards, Fellowships and Invited Talks

- Lab Director Trainee award, Mass Spectrometry Applications in Clinical Labs, Palm Springs-CA, January 21-26 2017.
- Lab Director Trainee award, Mass Spectrometry Applications in Clinical Labs, Palm Springs-CA, February 21-25 2016.
- Honours in PhD in Analytical Chemistry, University of Barcelona, April 2014.
- Young Researcher travel award in Recent Advances in Food Analysis (RAFA), Thermo Fisher Scientific, Prague-CZ, November 5-8, 2013.
- Formacio de Investigadores (FI) grant from Generalitat de Catalunya, Spain, June 2010-May 2013.
- Invited speaker at MS Food Day, Trento-Italy, 2013. Topic: Analysis of veterinary drugs using LC-MS/HRMS in an official food safety laboratory.
- Invited speaker at Jornada del Cicle de Conferencies, Laboratori Agroalimentari, Barcelona-Spain, 2012. Topic: Nous Avencos en L'Aplicacio de la Cromatografia de liquids acoblada a espectrometria de masses d' alta resolucio.
- Erasmus Mundus Masters grant, September 2008-February 2010.

Committees

- Erasmus Mundus Alumni Association member, 2010-present.
- Students representative at the Program Management Committee, EMQAL, 2008-2010.

Workshops/Trainings

- Advanced Imaging Mass Spectrometry workshop, Vanderbilt University, Nashville-TN, April 24-27 2018.
- Metabolomics workshop, Michigan Regional Comprehensive Metabolomics Core, Ann Arbor-MI, May 8-11 2017.
- Lipids and Lipidomics workshop, American Society of Mass Spectrometry, San Diego-CA, February 2015.
- Environmental Omics - Approaches and technologies for a deeper understanding, Institute for Environmental Assessment and Water Research (IDAEA), Barcelona-Spain, September 25-26, 2014.
- Training in Q Exactive Operations (Orbitrap Mass Spectrometer), Thermo Fisher Scientific Training Institute, Barcelona-Spain, January 17-19, 2012.
- Management of Intellectual Property Rights in Biotechnology workshop, Biotech Consortium India Ltd., Department of Biotechnology, India, Chennai-India, February 22-23, 2007.
- Training in handling Bioprocess Reactors, Central Leather Research Institute, Chennai-India, December 14-27, 2005.
- Molecular Biology and Biotechnology inplant training, Helix Life Science Laboratory, Chennai-India, December 2-8, 2004.

Key Publications

For full list of publications - [Google Scholar Profile](#)

Jayaraman, A., **Kumar, P***, Marin, S., de Atauri, P., Mateo, F., Thomson, T. M., ... & Cascante, M. (2018). Untargeted metabolomics reveals distinct metabolic reprogramming in endothelial cells co-cultured with CSC and non-CSC prostate cancer cell subpopulations. PLoS one, 13(2), e0192175 (co first authorship).

Kesarwani, P., Kant, S., Prabhu, A., **Kumar, P.**, Graham, S., Buelow, K., Wilson, G. D., Miller, C & Chinnaiyan, P.(2018). Tryptophan metabolism contributes to radiation-induced immune checkpoint reactivation in glioblastoma. Clinical Cancer Research, 24(15), 3632-3643.

S Graham, **P Kumar**, R Bahado-Singh, A Robinson, D Mann, B Green (2016), Novel Metabolite Biomarkers of Huntington's Disease as detected by High-Resolution Mass Spectrometry. Journal of proteome research 15 (5), 1592–1601.

Cowan, E., **Kumar, P.**, Burch, K. J., Grieve, D. J., Green, B. D., & Graham, S. F. (2016). Treatment of lean and diet-induced obesity (DIO) mice with a novel stable obestatin analogue alters plasma metabolite levels as detected by untargeted LC–MS metabolomics. Metabolomics, 12(7), 124.

P. Kumar, A. Rubies, F. Centrich, M. Granados, N. Cortes-Francisco, J. Caixach, R. Companyo, (2013) Targeted analysis with benchtop quadrupole-orbitrap hybrid mass spectrometer: Application to determination of synthetic hormones in animal urine, Anal. Chim. Acta, 780, 65-73.